1. Consider a program that maintains a list. The list supports the following commands:
   a. add e: insert an integer e at the end of the list.
   b. remove i: remove the element at index i from the list.
   c. print: print the content of the list.

   Write a program that accepts a positive integer N, followed by N commands. The program then performs each command in order.

Sample Program Output 1:

```
Enter N: 4
Enter command: add 123
Enter command: add 4
Enter command: add 56
Enter command: print
[123, 4, 56]
```

Sample Program Output 2:

```
Enter N: 8
Enter command: add 123
Enter command: add 4
Enter command: remove 0
Enter command: print
[4]
Enter command: add 56
Enter command: add 789
Enter command: remove 1
Enter command: print
[4, 789]
```

2. Create a text file named data.txt. Type some lines of words inside and save the file. Then, write a program that reads the text file, and show the number of lines in the file, the first line, and the last line in the file. Ensure that your program is saved in the same folder that contains data.txt.

If the content of data.txt is:

```
My Python
is
not
a snake
```

Then the output should be:

```
Line count: 4
First line: My Python
Last line : a snake
```

3. A text file named vehicles.txt is provided. Write a program to open vehicles.txt and count the number of the word “car” found in the file. The expected output should be 5.

4. A text file named numbers.txt is provided. It contains a list of integers with one integer per line. Write a program to find out the total number of integers, the largest, the smallest, the sum, and the average of the integers. Ensure that the output formatting is similar to the Expected Program Output below. Your program should still produce correct results even if the content of numbers.txt is changed.

Expected Program Output:

```
Count =  16
Largest =  9048 in line 9
Smallest =  885 in line 4
Sum =  82721
Average =  5170.0625
```

5. A text file named months.txt is provided. It has 12 lines with each line containing one of the months of the year. Write a program that asks the user for a letter and then displays the months containing that letter.

Sample Program Output 1:

```
Enter character to search: r
January, February, March, April, September, October, November,
December
```

Sample Program Output 2:

```
Enter character to search: k
No answer.
```

6. The keyboard in use on nearly all computers is known as the Qwerty keyboard since the letters in the top letter line read QWERTYUIOP. A word is called a Qwerty word if all its letters appear on the top letter line of the keyboard. Some examples are typewriter, repertoire, and treetop. Write a program that requests a word as input and then determines whether it is a Qwerty word. Use a Boolean-valued function named `isQwerty()` that evaluates the word.

Sample Program Output 1:

```
Enter a word: TRY
TRY is a Qwerty word.
```

Sample Program Output 2:

```
Enter a word: Poly
Poly is not a Qwerty word.
```